

GLOBAL FINANCIAL CRISIS 2008-09

Mridul Saggar
Professor of Practice (Economics)

questions & correspondence: mksaggar@iimk.ac.in

*This course material is for restricted use of the course participants alone
& strictly not for circulation*

What was the Global Financial Crisis (GFC)?

2



- GFC was a period of extreme stress in global financial markets and banking systems between mid 2007-mid 2009 that actually exploded into a full-scale global financial crisis in 2008.
- It was known as the Sub-prime crisis in its earlier phases, reflecting credit bubble to low-income homebuyers in the US.
- During the GFC, a downturn in the US housing market was a catalyst for a financial crisis that spread from the United States to the rest of the world through linkages in the global financial system. Many banks around the world incurred large losses and relied on government support to avoid bankruptcy.
- GFC reflected a complete market failure and the international financial system was on verge of a breakdown due to serial failure or near-failure of several commercial and investment banks, insurance companies, mortgage lenders and saving and loan associations.
- It resulted in the Great recession of 2007-09, which was the worst economic downturn since the Great Depression of 1929-39.

Seeds for the Origins of the Global Financial Crisis (GFC) ... (1)

3



Senators Gramm, Leach
& Bliley moving the
Financial Services
Modernization Bill

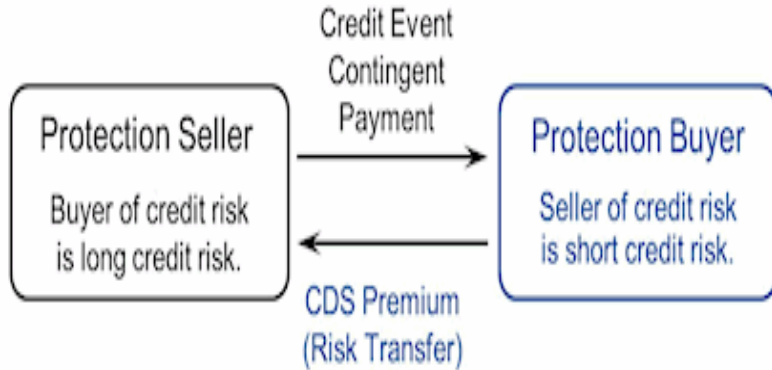
- **1992:** Congress passes legislation that requires government-sponsored mortgage giants Fannie Mae and Freddie Mac to devote a percentage of their lending to affordable housing. This led to an increase in the overall number of loans being pooled and securitized or sold as financial instruments to other investors. Two years later, J.P. Morgan introduces the first credit default swap (CDS), a credit derivative designed to act as a kind of insurance against defaults for investors. Over the next decade and a half, CDS become the most widely traded credit derivative product globally. The CDS market proves to be a major source of systemic risk to the U.S. financial system during the GFC.
- **1995-1997:** Subprime lending grew in the U.S. after the 1995 changes to the **Community Reinvestment Act**. The Act was originally passed to end discrimination against low-income borrowers. It allowed mortgage lenders to buy “subprime” securities to fulfill their affordable-housing lending obligations.
- **1999:** In September, Fannie Mae eased credit standards to encourage banks to extend loans to people whose credit scores were lower than what was required for conventional loans. On 12th November, **Financial Services Modernization Act (popularly known as Gramm-Leach-Bliley Act)** substantially repealed the **Banking Act of 1933 (popularly known as the Glass-Steagall Act)** that through its Section 20 provided for separation of commercial and investment banking. It paved the way for banks holding companies to set up subsidiaries to engage several para-banking activities, including underwriting and dealing in securities. The new Act paved the way for financial holding company structures over which Fed was provided supervisory powers.

Deregulation of the Financial Services increased Risks in the U.S.

4

Credit Default Swap (CDS)

The Basic Idea



The Protection Seller receives a periodic premium from the Protection Buyer in exchange for a contingent payment if there is a pre-defined Credit Event of a pre-specified Reference Entity.

The contingent payment is determined based on pre-specified settlement terms.

- Bank lobbyists backed repeal of the Glass-Steagall Act saying they needed this change to compete with foreign firms. They promised to only invest in low-risk securities to protect their customers. But steady erosion of credit discipline followed.
- In 2000, **the Commodity Futures Modernization Act** exempted credit default swaps (CDS) and other derivatives from regulations. This federal legislation overruled the state laws that had formerly prohibited this form of gambling. It specifically exempted trading in energy derivatives. Both bills were moved by Texas Senator Phil Gramm, Chairman of the Senate Committee on Banking, Housing, and Urban Affairs, whose wife was CFTC Chair and an Enron Board member and Enron had vested interest in these legislations because it wanted to use derivatives trading. Greenspan & Larry Summers supported these bills.

Seeds for the Origins of the Global Financial Crisis (GFC) ... (2)

5



Travelers Group CEO Sanford "Sandy" Weill (left) and Citicorp CEO John Reed (right) announcing the massive, historic \$70 billion merger between their two companies to form Citigroup Inc. on April 6, 1998 in NYC that heralded the start of an era of banks becoming large holding companies eventually scrapping Glass-Steagall Act

- In 2000-2001, Fed cut its policy interest rates after the **dot.com bubble** burst. This led to fresh credit bubble with easy mortgage lending.
- In April, the Securities Exchange Commission (SEC) relaxed the net capital rule which had capped broker-dealers and investment banks to a leverage (debt/ equity ratio) of 12:1. As part of the Consolidated Supervised Entities (CSE) Programme, SEC allowed firms with over US\$5 billion in assets to use their own internal mathematical models to measure risk. This effectively lifted the traditional 12:1 leverage ratio, allowing banks to leverage their capital an unlimited number of times.
- Bear Sterns, Lehman Brothers, Merrill Lynch, Goldman Sachs, Morgan Stanley increased the leverage to ~30 times and thus, failed to survive as independent I-banks.
- At the same time, SEC relieved I-banks of standard broker-dealer constraints, major investment banks rapidly expanded their balance sheets by funding highly complex, mortgage-backed securities (MBS) and collateralized debt obligations (CDOs) with short-term borrowing.

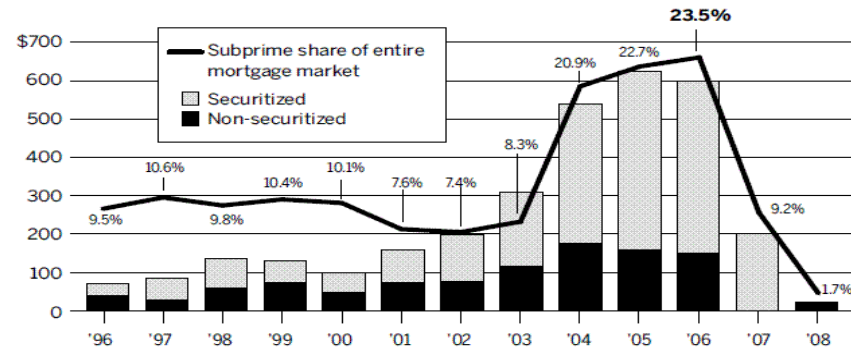
What is the Sub-prime Lending that was led to the Sub-prime Crisis that later turned into the Global Financial Crisis (GFC)?

6

Subprime Mortgage Originations

In 2006, \$600 billion of subprime loans were originated, most of which were securitized. That year, subprime lending accounted for 23.5% of all mortgage originations.

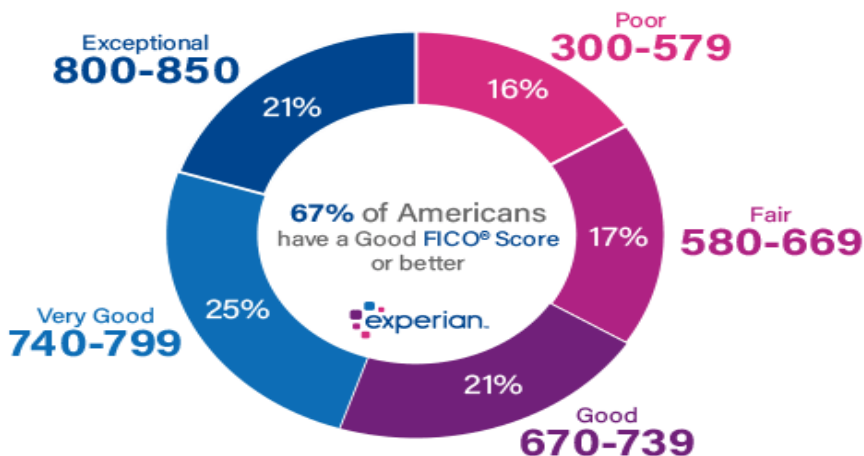
IN BILLIONS OF DOLLARS



NOTE: Percent securitized is defined as subprime securities issued divided by originations in a given year. In 2007, securities issued exceeded originations.

SOURCE: Inside Mortgage Finance

- The predatory lending practices adopted by the US banks targeting low-income homebuyers, excessive risk-taking by global financial institutions was the source of the GFC.
- **Predatory lending:** No internationally agreed legal definitions but a 2006 audit report from the office of inspector general of the US FIDC defines it as "imposing unfair and abusive loan terms on borrowers". Though the Report did not define it further, practices such as high interest rates, high fees and aggressive tactics and deception to get borrowers to take loans they can't afford is usually considered predatory. Balloon payments are often used.
- Subprime lending are loan arrangements with borrowers with poor credit history who are lent at rates well above the prime rate. In the US, credit history is represented in Fair, Isaac Corporation (FICO) scores and categorized as follows:
 - 800 to 850: Excellent, i.e. low risk;
 - 740 to 799: Very good credit history; additional loans get approved easily
 - 670 to 739: Good, i.e. acceptable or lower risk borrowers
 - 580 to 669: Fair, i.e. higher risk and subprime;
 - 300 to 579: Poor, i.e. high risk and you need to improve credit scores before you can get new credit

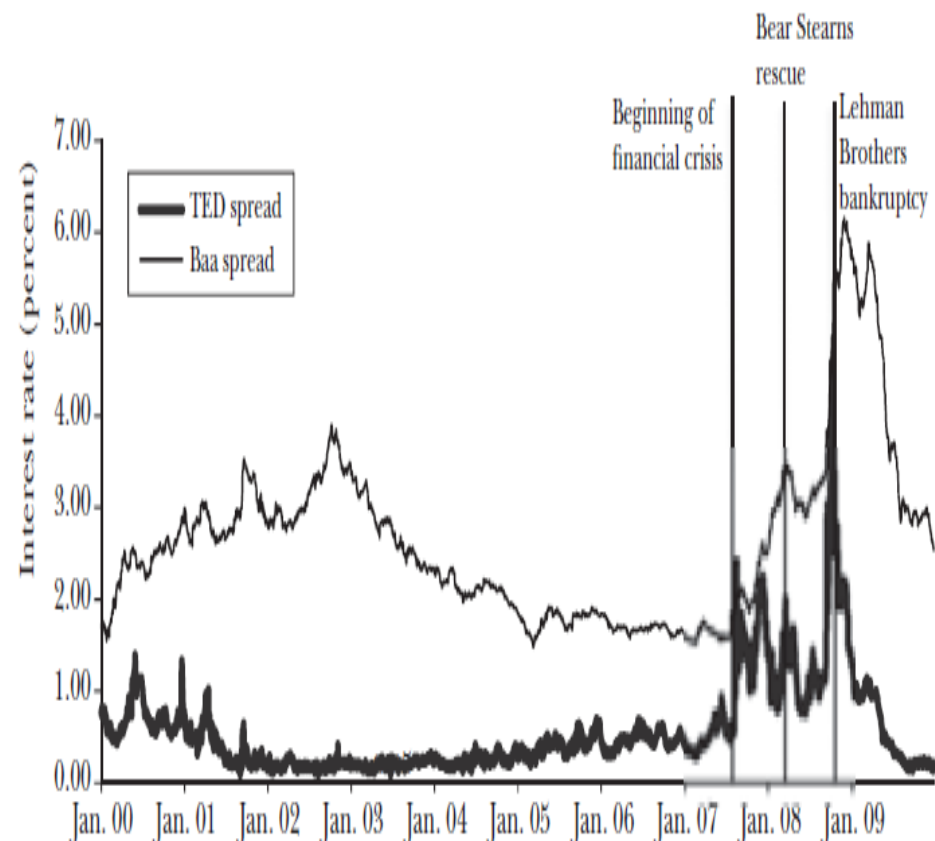


With Deregulation of the Financial Services & resultant increased risks

Credit Spreads widened alarmingly

7

Credit Spreads 2000-2009



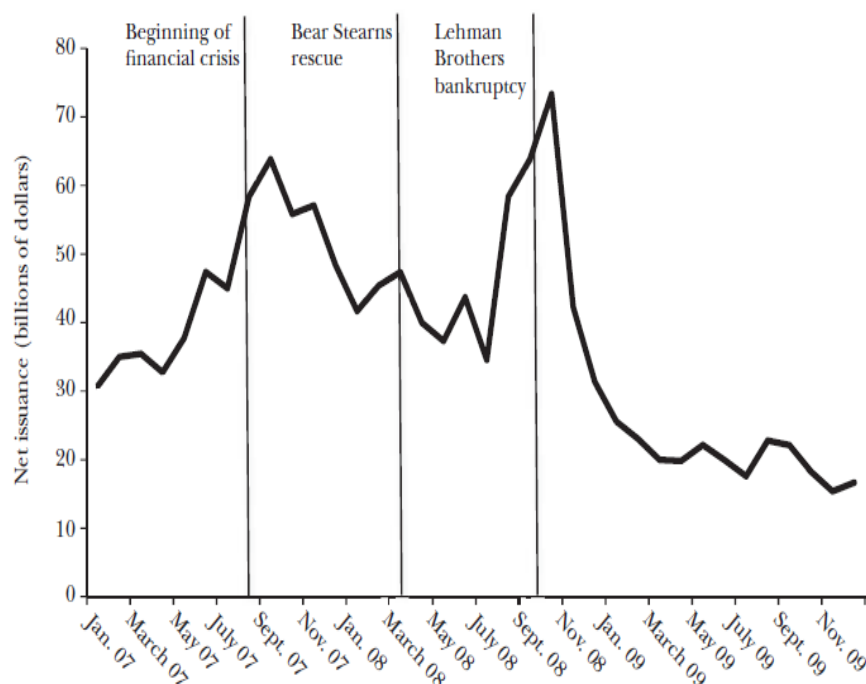
Source: Mishkin (2011), JEP

- **What is TED Spread?** It is the difference between the three-month LIBOR and the three-month Treasury bill rate. It is commonly used as a measure of credit risk, as U.S. Treasury bills are seen as risk-free. The TED spread often widens in periods of economic crisis, as the default risk widens; the spread narrows when the economy is more stable and defaults are less of a risk.
- **What is the Baa Spread?** It is the spread between Moody's seasoned corporate bonds with Baa (medium rating) yield minus the 10-year Treasury constant maturity yield. It is again a measure of credit risk.
- **What is the US VIX?** It is 30-day expected volatility of the U.S. stock market, derived from real-time, mid-quote prices of S&P 500 Index call and put options.
- **What is MOVE?** It is Merrill Lynch Option Volatility Estimate for the bond market (equivalent of the equity market's VIX). It measures expected, or implied, volatility in the U.S. Treasury market by tracking the prices of 1-month options on 2-year, 5-year, 10-year, and 30-year USTs.

Securitization obfuscated risks and led to the Global Financial Crisis (GFC)

8

C: Asset-Backed Commercial Paper Issuance



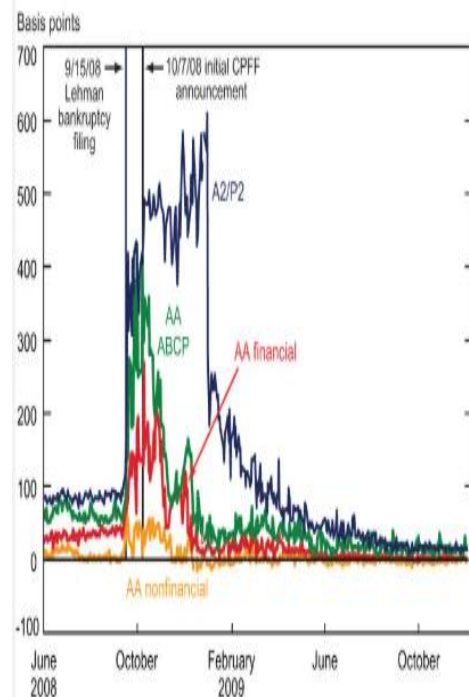
Source: Mishkin (2011), JEP

- **Securitization:** Hedge funds and others sold mortgage-backed securities (MBS), collateralized Debt obligations (CDOs) and other derivatives. A mortgage-backed security is a financial product whose price is based on the value of the mortgages that are used for collateral. Once you get a mortgage from a bank, the bank sells the loan on the secondary market. The hedge fund then bundles your mortgage with a lot of other similar mortgages.
- They used computer models to figure out what the bundle is worth based on several factors. These included the monthly payments, the total amount owed, the likelihood you will repay, and future home prices.
- The hedge fund then sold the MBS to investors taking liability off-balance sheet. It can then make new loans with the money it received. It may still collect your payments, but it sends them along to the hedge fund, which sends it to its investors. So, the risk was transferred from bank to hedge fund and from hedge fund to investor.
- Investors did not worry because Credit Default Swaps (CDS) offered them insurance. But these CDS were again owned by financial institutions like Bear sterns, Citibank and Lehman Brothers

Securitization obfuscated risks and led to the Global Financial Crisis (GFC)

9

One-Month Commercial Paper Spread



Sources: Board of Governors of the Federal Reserve System; Bloomberg L.P.

Note: The chart shows the spread between the one-month commercial paper rate and the overnight index swap rate.

- Another instrument used was Asset Backed Commercial Papers (ABCP). Asset-backed commercial paper is like traditional commercial paper, in that it is issued with maturities of one year or less (typically less than 270 days) and is highly rated. The difference between ABCP and regular CP is that instead of being an unsecured promissory note representing an obligation of the issuing company, ABCP is backed by securities. Therefore, the perceived quality of the ABCP depends on the underlying securities, typically an SPV so that it is bankruptcy remote.
- Unlike standard commercial paper, ABCP is backed (collateralized) by a pool of assets, such as auto loans, credit card receivables, or trade receivables and the quality of this collateral backing was at times dubious, serving to obfuscate risks on the balance sheets.
- Banks typically used ABCP to move assets off their balance sheets while still maintaining liquidity lines for their clients. ABCPs were also used by the shadow banking system (NBFIs) to fund long-term, illiquid assets using very short-term debt. Investors frequently rolled over these securities every few days.
- In Aug 2007, doubts about the quality of subprime mortgages hidden in these asset pools led investors to stop buying ABCPs. This market freeze stranded issuers with billions in illiquid, un-rollable debt. Fed instituted emergency lines - the Asset-Backed Commercial Paper Money Market Mutual Fund Liquidity Facility (AMLF) and the Commercial Paper Funding Facility (CPFF)

How the Global Financial Crisis (GFC) unfolded...(1)

10

- Tuesday, April 3, 2007: **New Century Financial**, which specialized in sub-prime mortgages, files for Chapter 11 bankruptcy protection and cuts half of its workforce.
- Thursday, May 17, 2007: Federal Reserve Chairman Ben Bernanke says growing number of mortgage defaults will not seriously harm the US economy.
- Wednesday, June 2007: Two **Bear Stearns-run** hedge funds with large holdings of subprime mortgages run into large losses and are forced to dump assets. The trouble spreads to major Wall Street firms such as Merrill Lynch, JPMorgan Chase, Citigroup and Goldman Sachs which had loaned the firms money.
- July 2007: Investment bank Bear Stearns tells investors they will get little, if any, of the money invested in two of its hedge funds after rival banks refuse to help it bail them out.
- Thursday, August 9, 2007: Investment bank **BNP Paribas** tells investors they will not be able to take money out of two of its funds because it cannot value the assets in them, owing to a "complete evaporation of liquidity" in the market. The European Central Bank pumps €95bn (£63bn) into the banking market to try to improve liquidity. It adds a further €108.7bn over the next few days. The US Federal Reserve, the Bank of Canada and the Bank of Japan also begin to intervene.
- Friday, August 17, 2007: The Fed cuts the rate at which it lends to banks by half of a percentage point to 5.75%, warning the credit crunch could be a risk to economic growth.
- Tuesday, September 4, 2007: The rate at which banks lend to each other rises to its highest level since December 1998. The so-called LIBOR rate is 6.7975%, way above the Bank of England's 5.75% base rate; banks either worry whether other banks will survive, or urgently need the money themselves.

How the Global Financial Crisis (GFC) unfolded...(2)

11

- Thursday, September 13, 2007: The BBC reveals **Northern Rock** has asked for and been granted emergency financial support from the Bank of England, in the latter's role as lender of last resort. Northern Rock relied heavily on the markets, rather than savers' deposits, to fund its mortgage lending. The onset of the credit crunch has dried up its funding.
- Friday, September 14, 2007: Depositors withdraw £1bn from Northern Rock in what is the biggest run on a British bank for more than a century. They continue to take out their money until the government steps in to guarantee their savings.
- Tuesday, September 18, 2007: The US Federal Reserve cuts its main interest rate by half a percentage point to 4.75%.
- Wednesday, September 19, 2007: After previously refusing to inject any funding into the markets, the Bank of England announces that it will auction £10bn.
- Monday, October 1, 2007: Swiss bank UBS is the world's first top-flight bank to announce losses - \$3.4bn - from sub-prime related investments. The chairman and chief executive of the bank step down. Later, banking giant Citigroup unveils a sub-prime related loss of \$3.1bn. A fortnight on, Citigroup is forced to write down a further \$5.9bn. Within six months, its stated losses amount to \$40bn.
- Tuesday, October 30, 2007: Merrill Lynch's chief resigns after the investment bank unveils a \$7.9bn exposure to bad debt.
- Thursday, December 6, 2007: US President George W Bush outlines plans to help more than a million homeowners facing foreclosure. The Bank of England cuts interest rates by a quarter of one percentage point to 5.5%.
- Thursday, December 13, 2007: The US Federal Reserve co-ordinates an unprecedented action by five leading central banks around the world to offer billions of dollars in loans to banks. BoE calls it an attempt to "forestall any prospective sharp tightening of credit conditions." The move succeeds in temporarily lowering the rate at which banks lend to each other.

How the Global Financial Crisis (GFC) unfolded...(3)

12

- Monday, December 17, 2007: The central banks continue to make more funding available.
- There is a \$20bn auction from the US Federal Reserve and, the following day, \$500bn from the European Central Bank to help commercial banks over the Christmas period.
- Wednesday, December 19, 2007: Ratings agency Standard and Poors downgrades its investment rating of a number of so-called monoline insurers, which specialise in insuring bonds.
- Monday, January 21, 2008: Global stock markets, including London's FTSE 100 index, suffer their biggest falls since 11 September 2001.
- Tuesday, January 22, 2008: The US Fed cuts rates by three quarters of a percentage point to 3.5% - its biggest cut in 25 years - to try and prevent the economy from slumping into recession. It is the first emergency cut in rates since 2001. Stock markets around the world recover the previous day's heavy losses.
- Thursday, February 7, 2008: US Federal Reserve boss Ben Bernanke adds his voice to concerns about monoline insurers, saying he is closely monitoring developments "given the adverse effects that problems of financial guarantors can have on financial markets and the economy." The Bank of England cuts interest rates by a quarter of one percent to 5.25%.
- Friday, February 8, 2008: In the UK, the latest CML figures show the number of homes repossessed in the UK rose to 27,100 in 2007, its highest level since 1999.
- Sunday, February 10, 2008: Leaders from the G7 group of industrialised nations say worldwide losses stemming from the collapse of the US sub-prime mortgage market could reach \$400bn.
- Sunday, February 17, 2008: British government nationalizes Northern Rock.

How the Global Financial Crisis (GFC) unfolded...(4)

13

- Friday, Mar 7, 2008: In its biggest intervention yet, the Federal Reserve makes \$200bn of funds available to banks and other institutions to try to improve liquidity in the markets.
- Sunday, Mar 16, 2008: Bear Stearns is bought by J.P. Morgan Chase in a deal orchestrated by and backed by the US government, following a sharp decline in shares and a collapse in the confidence in the company.
- Wednesday, Apr 2, 2008: Moneyfacts, which monitors financial products, says 20% of mortgage products have been withdrawn from the UK market in the previous seven days. Five days later the 100% mortgage disappears when Abbey withdraws the last home loan available without a deposit.
- Tuesday, Apr 8, 2008: The IMF warns that potential losses from the credit crunch could reach \$1 trillion and may be even higher. It says the effects are spreading from sub-prime mortgage assets to other sectors, such as commercial property, consumer credit, and company debt.
- Thursday, Apr 10, 2008: The Bank of England (BoE) cuts interest rates by a quarter of one percent to 5%.
- Tuesday, Apr 15, 2008: Confidence in the UK housing market falls to its lowest point in 30 years in March, according to the Royal Institution of Chartered Surveyors, because of the "unique liquidity blight." But it does add that the situation is good news for buyers with large deposits who can buy property that was previously out of reach.
- Monday, Apr 21, 2008: The BoE announces details of an ambitious £50bn plan designed to help credit-squeezed banks by allowing them to swap potentially risky mortgage debts for secure government bonds.
- Tuesday, Apr 22, 2008: Royal Bank of Scotland announces a plan to raise money from its shareholders with a £12bn rights issue - the biggest in UK corporate history. The firm also announces a write-down of £5.9bn on the value of its investments between April and June - the largest write-off yet for a British bank.

How the Global Financial Crisis (GFC) unfolded...(5)

14

- Wed, Apr 30, 2008: The first annual fall in house prices for 12 years is recorded by Nationwide. Prices were 1% lower in April compared to a year earlier after a "steep decline" in home buying over the previous six months. Later in the week, figures from the UK's biggest lender Halifax, show a 0.9% annual fall for April.
- Thurs, May 22, 2008: Swiss bank UBS, one of the worst affected by the credit crunch, launches a \$15.5bn rights issue to cover some of the \$37bn it lost on assets linked to US mortgage debt.
- In a crackdown on mortgage frauds worth \$1 bn, FBI arrests 406 people, including brokers and housing developers. Separately, criminal charges were levied on two former Bear Stearns workers linked with the collapse of two hedge funds sub-prime mortgages, as they knew of the funds' problems but did not disclose them to investors, who lost a total of \$1.4bn.
- Wed, June 25, 2008: Barclays announces plans to raise £4.5bn in a share issue to bolster its balance sheet. The Qatar Investment Authority agreed to invest £1.7bn in the British bank, giving it a 7.7% share in the business. A number of other foreign investors increase their existing holdings.
- Fri, July 11, 2008: American Federal regulators seize IndyMac Bank after it succumbs to the pressure of tighter credit, tumbling home prices and rising foreclosures. IndyMac is the largest thrift ever to fail in the United States. Barrel of oil hits a record price of \$147.5.
- Mon, July 14, 2008: Financial authorities step in to assist two largest US lenders, Fannie Mae and Freddie Mac who guaranteed \$5 trillion of home loans, as authorities agreed despite their plummeting share prices they cannot be allowed to fail.
- Thurs, July 31, 2008: UK house prices declined 8.1% - their biggest annual fall since the Nationwide began its housing survey in 1991, a decline of 8.1%. Meanwhile, HBOS reveals that profits for the H12008 sank 72% to £848m, while bad debts rose 36% to £1.31bn as customers failed to repay loans.

How the Global Financial Crisis (GFC) unfolded...(6)

15

- Monday, August 4, 2008: HSBC warns that conditions in financial markets are at their toughest "for several decades" after suffering a 28% fall in half-year profits. HSBC has among the heaviest exposures to the troubled US housing and credit markets. Amongst top European banks
- Saturday, August 9, 2008: Investment bank BNP Paribas tells investors they will not be able to take money out of two of its funds because it cannot value the assets in them, owing to a "complete evaporation of liquidity" in the market. The European Central Bank pumps €95 billion in to the banking market to try to improve liquidity. It adds a further €108.7 billion over the next few days.
- Monday, September 1, 2008: Official figures from the Bank of England show a slump in approved mortgages for July. Meanwhile, while the pound falls to record lows of 81.21 pence against the euro and two-year lows of \$1.80.
- Tuesday, September 2, 2008: In an effort to kick-start the UK housing market the Treasury announces a one-year rise in stamp duty exemption, from £125,000 to £175,000. OECD forecasts that the UK will be in a full-blown recession for UK.
- Wednesday, September 3, 2008: The European Central Bank (ECB) cuts growth forecast for 2009 to 1.2% from 1.5%, leaves interest rate unchanged at 4.25%. Next day the BoE kept rates unchanged.
- Sunday, September 7, 2008: Mortgage lenders Fannie Mae and Freddie Mac - which account for nearly half of the outstanding mortgages in the US - are rescued by the US government. Treasury Secretary Henry Paulson says the two firms' debt levels pose a "systemic risk" to financial stability and that, without action, the situation would get worse. At the same time, in the UK, the Nationwide announces it will merge with two smaller rivals, the Derbyshire and Cheshire Building Societies.
- Tue, Sept 9, 2008: More bad news emerges for the UK economy as the ONS reveal manufacturing output fell by 0.2% between June and July, raising a real fear of recession.

How the Global Financial Crisis (GFC) unfolded...(7)

16

- Wed, Sept 10, 2008: The US government seizes Fannie Mae and Freddie Mac. European Commission, which warns that the UK, Germany and Spain will go into recession by the end of the year.
- Thurs, Sept 11, 2008: Lehman Brothers announces it is actively looking to be sold after reporting US\$4 billion in losses.
- Fri, Sept 12, 2008: With Lehman Brothers facing collapse, Treasury struggles to find taker for the distressed investment bank.
- Sat, Sept 13, 2008: Teams of bankers flood the New York Federal Reserve building for the weekend to explore options for Lehman. Bank of America and Barclays lead the list for potential buyers.
- Sun, Sept 14, 2008: Talks at the New York Federal Reserve continues. Barclays pulls out of the bidding for Lehman as BoE indicated to them they would not approve the deal. Bank of America turns its attention to Merrill Lynch, saying it will buy it for \$29 per share. Lehman Brothers tell that it will file for Chapter 11 bankruptcy after the Fed refuse to bail it out. J.P. Morgan provides the broker dealer unit of Lehman Brothers with \$138 billion to settle securities transactions.
- Mon, Sept 15, 2008: Bank of America agrees to a \$50 billion rescue package for Merrill Lynch. Lehman files for bankruptcy; the largest bankruptcy filing in the history of the US at \$639 billion. FTSE 100 slumps 4%. US officials agree to put together a \$20 billion lifeline bid for insurance giant AIG as the Dow Jones plunges 504 points to close at 10,917.51
- Tue, Sept 16, 2008: Asian markets plummet starting with Nikkei falling 570 points. Goldman Sachs report 70% drop in profits. Barclays agree to take Lehman's US assets. US Treasury announces a US\$85 billion bailout package for AIG.
- Wed, Sept 17, 2008: LIBOR hits 7-year high. FTSE closes below 5000 points for the first time since May 2005, even as BoE extends its special liquidity scheme. Morgan Stanley looks to merge with Wachovia. Russian stock markets shut down for 2-days

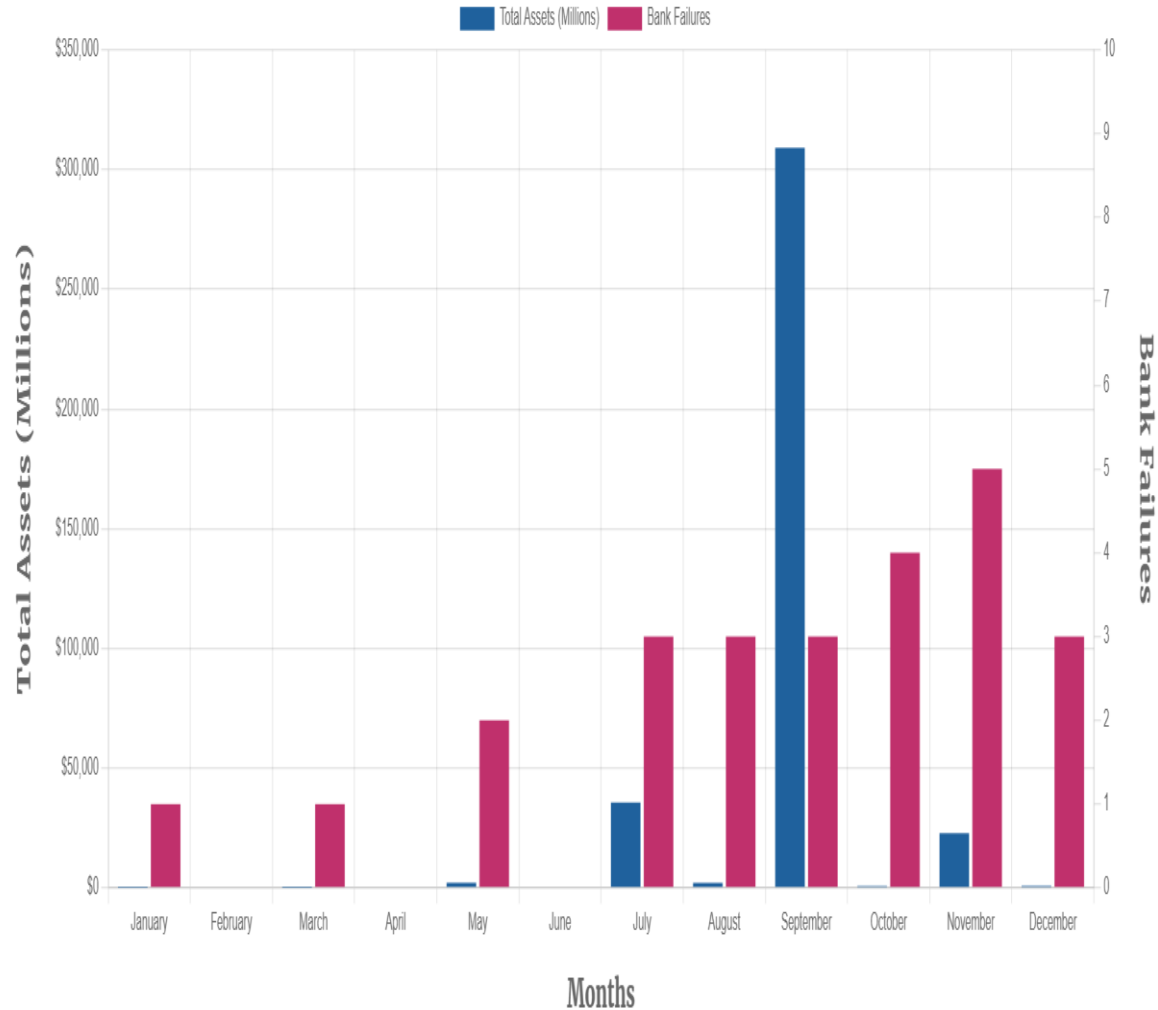
How the Global Financial Crisis (GFC) unfolded...(8)

17

- Thurs, Sept 18, 2008: Gold rises to 6-week high. The Dow Jones Industrial average plunges 504 points. Global central banks inject US\$180 bn in liquidity. Fed cuts policy rate 50 bps to 4.75%. U.K. FSA bans short selling of bank shares.
- Frid, Sept 19, 2008: Global stock markets stage recovery led by Asia.
- Sat, Sept 20, 2008: Henry Paulson announces US\$700 billion TARP bailout package
- Mon, Sept 22, 2008: Morgan Stanley and Goldman Sachs give up their status as investment banks and become traditional commercial banks. Nomura takes over Lehman's Asia operations.
- Frid, Sept 26, 2008: America's biggest savings-and-loan company, Washington Mutual, is seized by federal regulators and sold to J.P. Morgan for \$ 1.9 billion in a deal that sends shockwaves through Wall Street and Main Street alike. WaMu thus becomes the largest thrift failure with \$307 billion in assets.
- Thurs, Oct 2, 2008: The US Senate approves the bailout. Congress passes the \$700-billion TARP
- Fri, Oct 10, 2008: A global rout starts in Asia as recession fears deepen, with Japan's Nikkei index falling almost 10%, its biggest drop in 20 years. FTSE also crashes 10% and Dow plunges 8%. Massive bailouts follow, policy rate cuts and QEs followed.
- Mon, 1 Dec 2008: The US recession is confirmed by the NBER panel including economists from Stanford, Harvard and MIT.
- Sun, 7 Dec 2008: HSBC Holdings created a \$5 billion (3.4 billion pound) global working capital fund for SMEs. Indian PM Manmohan Singh unveiled a Rs 3 tn (\$60 billion) spending plan, four times more than expected, to bolster an economy buffeted by recession and a terrorist attack. RBI had cut interest rates for the third time in less than two months on the previous day.
- Global Financial Regulatory reforms followed the waning of the GFC. In the U.S. the Dodd–Frank Wall Street Reform and Consumer Protection Act, commonly referred to as Dodd–Frank was enacted on July 21, 2010.

Major US Bank Failures in 2008 during Global Financial Crisis (GFC)

18



- On May 9: ANB Financial with assets of US\$1.8 bn failed and Pulaski Bank & Trust Co agreed to assume the non-brokered insured deposits of US\$ 213 mn.
- On July 11: IndyMac Bank failed with \$32 bn in assets and US\$19.1 bn of non-brokered insured deposits were transferred to IndyMac Federal Bank under FIDC trusteeship
- On July 25: First National Bank of Nevada failed with US\$3.4 bn in assets and Omaha Bank assumed all assets
- On Sept. 15: Lehman Brothers filed for Bankruptcy with US\$619 bn in debt.
- On Sept. 25: Washington Mutual bank failed with assets of US\$307 bn & supported y FIDC, JP Morgan assumed US\$188 bn of its deposits

Troubled Asset Relief Program (TARP)

19



- In Sept. 2008, the US financial markets had frozen following the collapse of Lehman and troubles in Fannie Mae, Freddie Mac and AIG.
- As situation was spiraling out of control, Treasury Secretary **Henry Paulson** (formerly Chairman & CEO of Goldman Sachs) pioneered the **Troubled Asset Relief Program (TARP)**. It was signed into law by President George W. Bush on October 3, 2008, with the passage of the **Emergency Economic Stabilization Act**.
- TARP's original aim was to increase the liquidity of the money markets and secondary mortgage markets by purchasing the mortgaged-backed securities (MBS) and through that, reduce the potential losses of the institutions that owned them
- Later, TARP's aim was modified to allow the government to buy equity in banks and other financial institutions. TARP initially gave the Treasury purchasing power of \$700 bn. The Dodd-Frank Wall Street Reform and Consumer Protection Act (Dodd-Frank Act) later reduced the authorization to \$475 bn.
- The U.S. government bought preferred stock in eight banks: Bank of America/Merrill Lynch, Bank of New York Mellon, Citigroup, Goldman Sachs, J.P. Morgan, Morgan Stanley, State Street, and Wells Fargo. The banks were required to give the government a 5% dividend that would increase to 9% in 2013, encouraging banks to buy back the stock within five years.
- From the inception till Oct 3, 2010 (the deadline for extending funds), \$245 billion was used to stabilize banks, \$27 billion went to programs to increase credit availability, \$80 billion went to the U.S. auto industry (specifically, GM and Chrysler), \$68 billion was used to stabilize AIG, and \$46 billion went to foreclosure-prevention
- TARP booked US\$15.3 billion in profits for taxpayers recovering \$441.7 billion from \$426.4 billion invested.

The Dodd-Frank Act, 2010

20

- The **Dodd–Frank Wall Street Reform and Consumer Protection Act**, commonly referred to as the Dodd–Frank Act was a U.S. Federal law enacted under Barack Obama’s Presidency on July 21, 2010 to promote the financial stability of the U.S. by improving accountability and transparency in the financial system, to end “too big to fail (TBTF)” to protect the American taxpayer by ending bailouts and to protect consumers from abusive financial services practices, and for other purposes.

Some **Key Provisions** of the Act:

- Created Financial Stability Oversight Council (FSOC) to identify fragilities & for regulatory coordination.
- Orderly Liquidation Authority (OLA) to wind down large, complex financial institutions.
- Consumer Financial Protection Bureau (CFPB): An independent agency for consumer protection by regulating consumer financial products and services.
- **Volcker Rule:** Restricting banks from speculative proprietary trading and limiting their investments in hedge funds and private equity funds to minimize risk to depositors and taxpayers. They were complemented by **Bonus rules/** incentive compensation rules, to regulate compensation structures so as not to incentivize excessive risk-taking.
- Derivatives reforms for trading them on exchanges/ electronic platforms and clearing through CCPs.
- Mandated regular stress tests of banks and increased capital requirements for banks
- Established overseeing credit rating agencies through SEC to provide meaningful and reliable ratings.
- Strengthen & expand the Whistleblower Program with rewards and protection to individuals exposing inappropriate behavior.

In 2018, the Economic Growth, Regulatory Relief, and Consumer Protection Act rolled back some Dodd-Frank provisions, particularly easing regulations for smaller and mid-sized bank

Some Lessons from the GFC

21



Mark Carney, earlier BoC & BoE Governor pushed TBTF reforms as FSB Chairman during 2011-18; is now Canada's PM



Trump signing executive orders in Feb 2005 expanding powers over independent regulatory agencies

- Asset price booms and busts keep repeating but housing bubbles especially need to be tracked and pricked
- Weak regulation can cause crisis and lead to huge output losses and sufferings. GFC was not a liquidity crisis.
- On the contrary, creation of excess liquidity fuelled the crisis
- Financial innovations need to be watched closely and curbed if necessary
- A clear idea on which firm may be systemically important for bailout is important.
- 'Too Big To Fail (TBTF)' problem must be addressed
- Capital and Liquidity buffers for banks need to be strengthened.
- Misaligned internal incentives (excessive bonuses) in financial firms create perverse incentives and pose risks
- Financial World is integrated and require global coordination in many respects
- Insolvency, LOLR and bailout policies need to be strengthened.